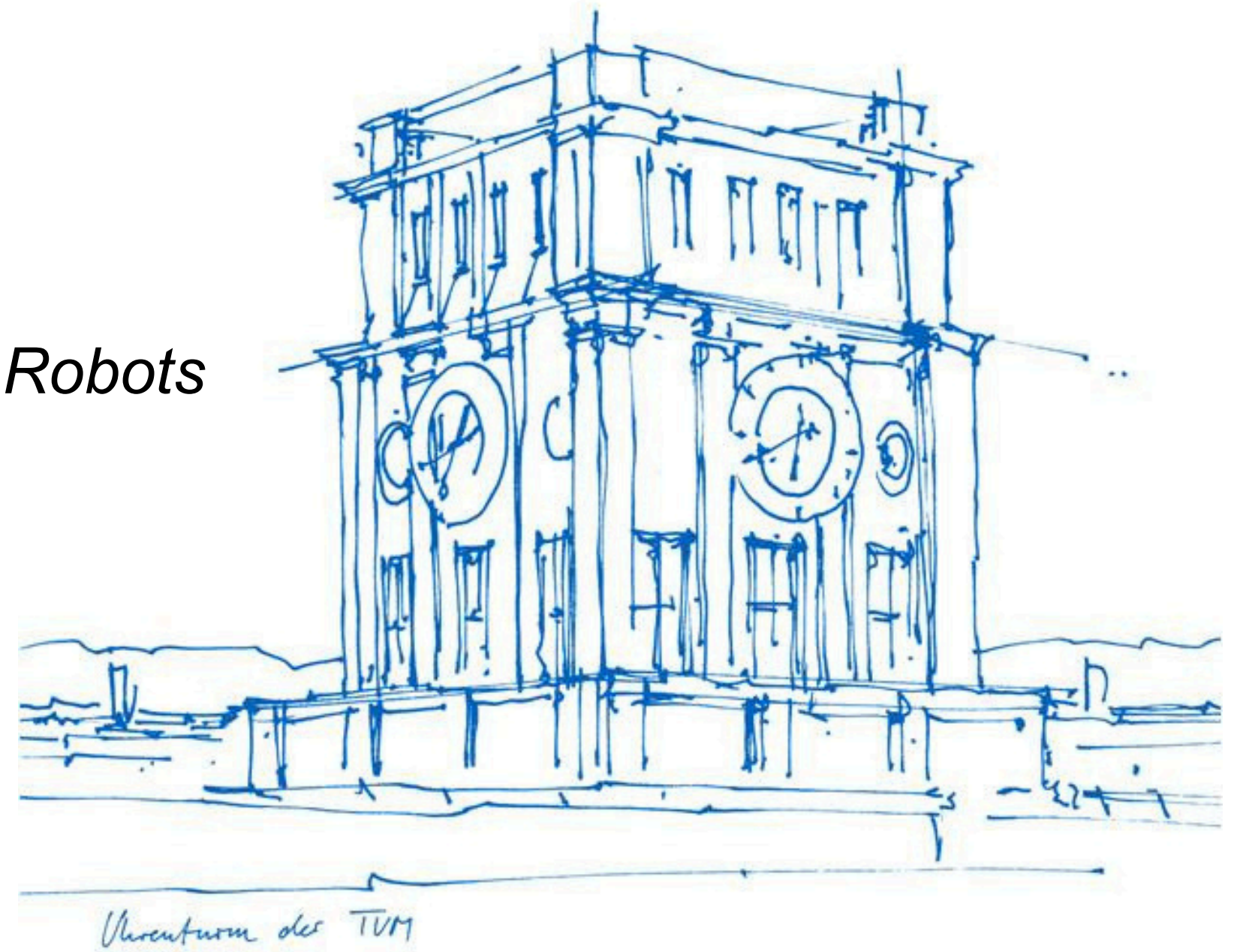


Modelling and control of legged robots - Final Presentation

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Obstacle Track Locomotion Simulation for Legged Robots

Technische Universität München
Munich, 23. July 2026

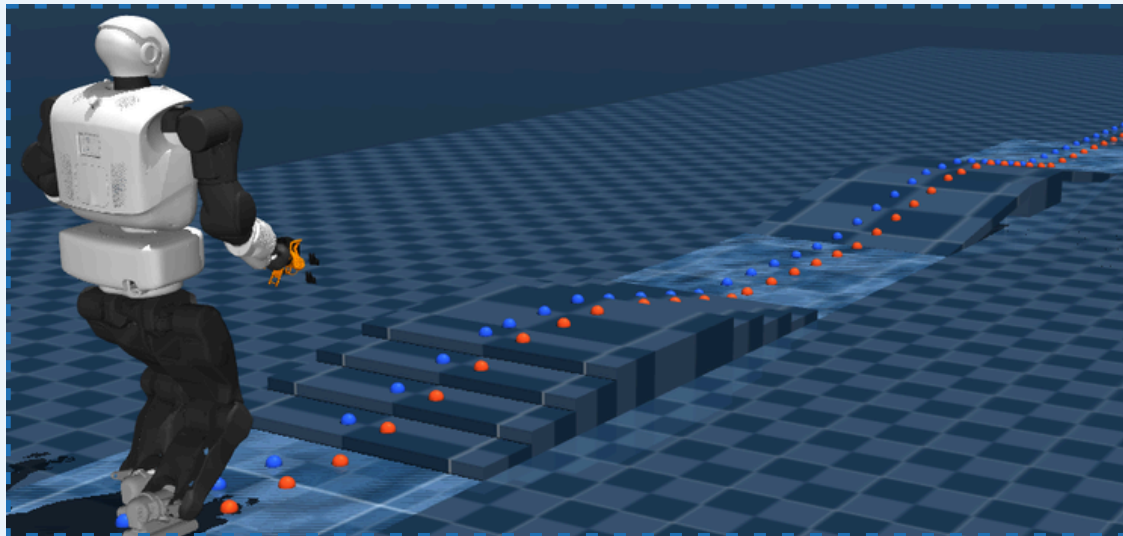


Track Setup

Three-Level Progressive Terrain Environments

LEVEL 1

Fully known · 1-D

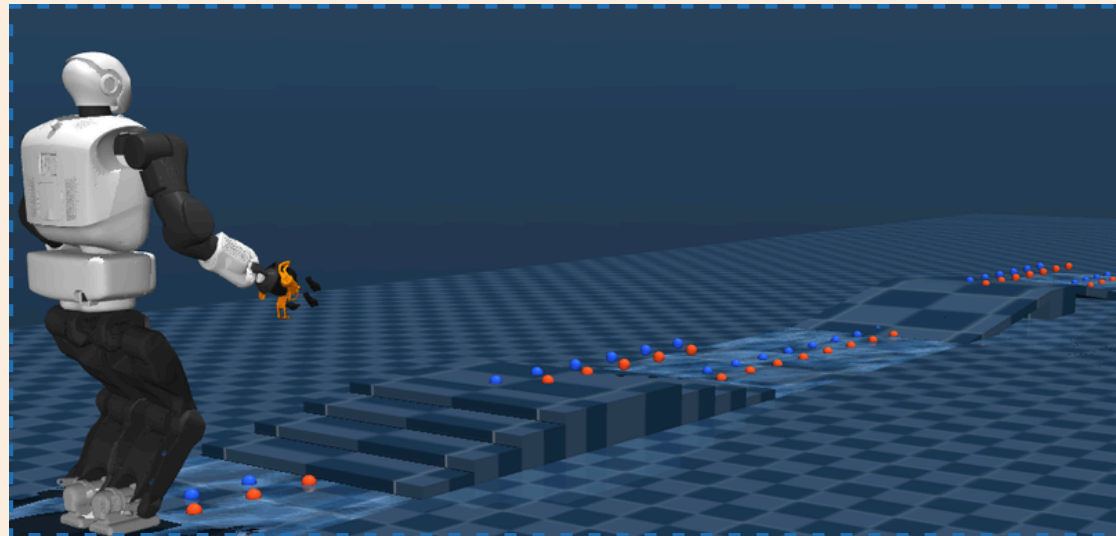


- Fixed obstacle positions along the world (x- axis)
- Full terrain information given in advance
- Straight-line 1-D walking path
- No online replanning required

Baseline — Verify basic terrain walking

LEVEL 2

Sensing · 1-D

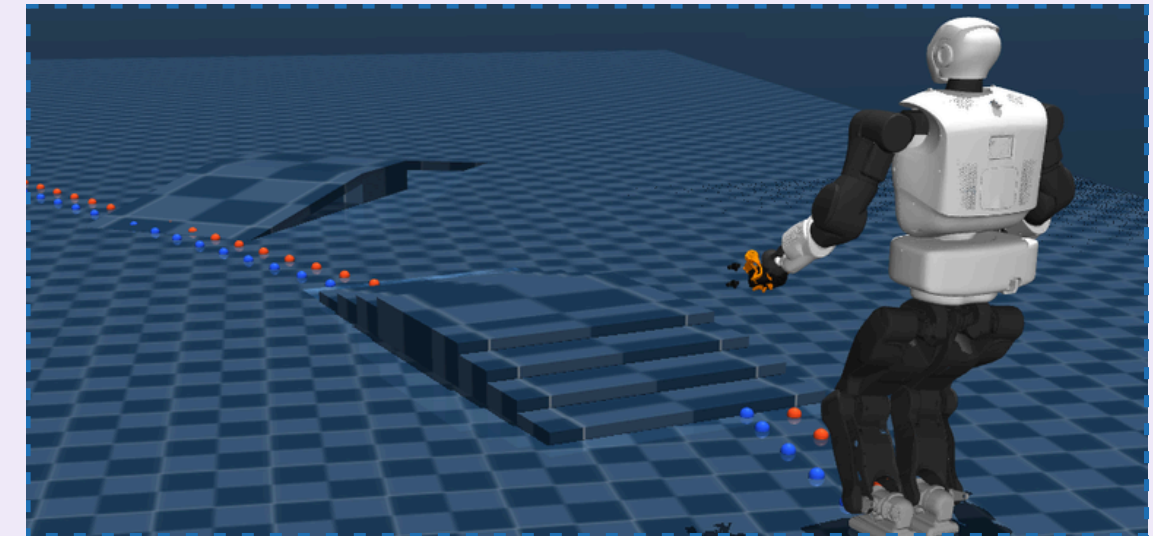


- Only nominal obstacle entries are known at the start
- True entry randomized along x: ± 0.50 m
- Sensor range 0.70 m
- Footsteps replan after each detection

Sense → Detect → Online replanning

LEVEL 3

Turning · 2-D

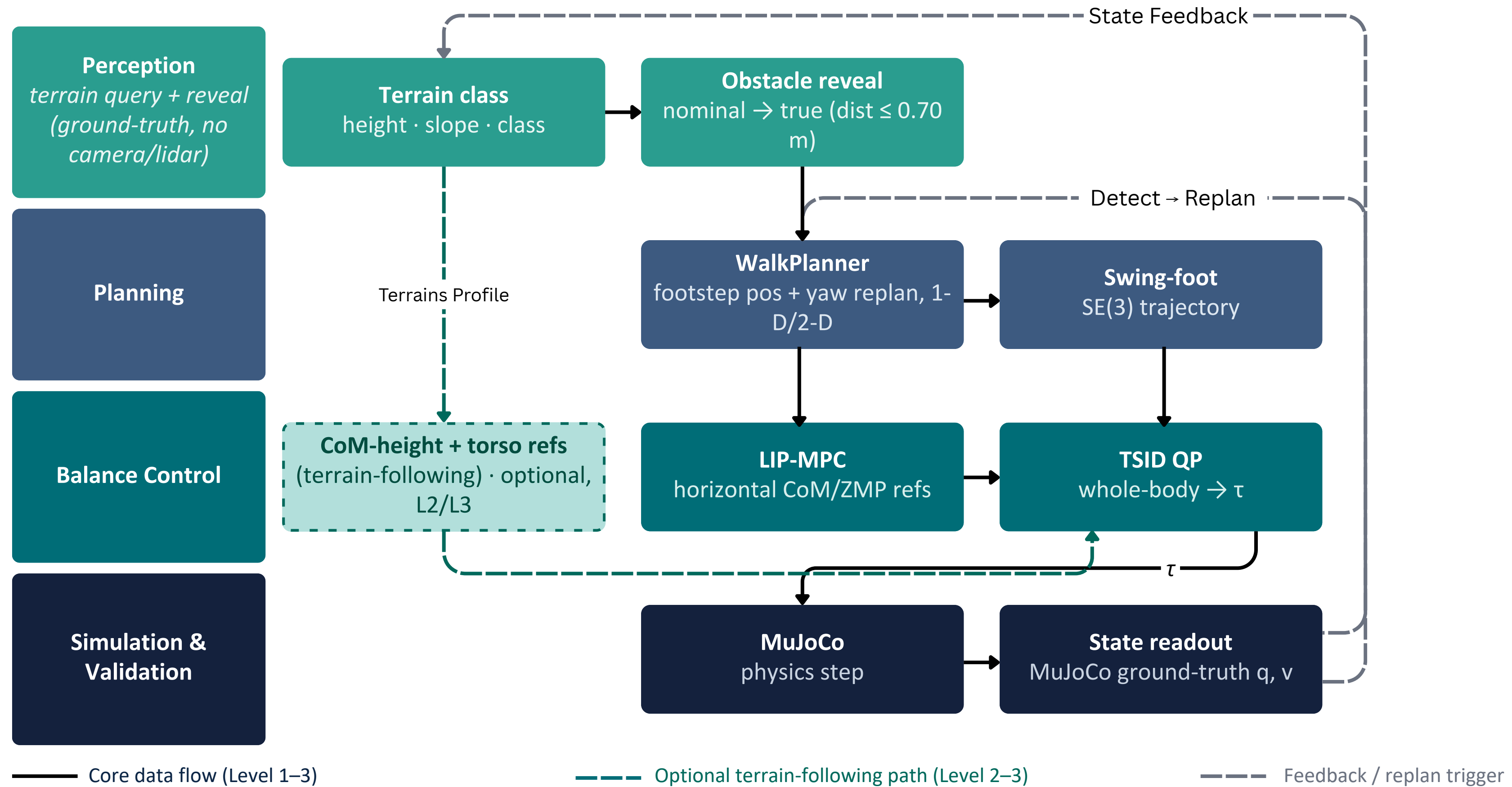


- Obstacles have true (x, y, yaw) poses
- x/y randomized ± 0.50 m · yaw ± 0.40 rad ($\approx \pm 23^\circ$)
- 2-D replanning with turning
- Yaw-aware footstep placement

2-D planning + Turning + Online replanning

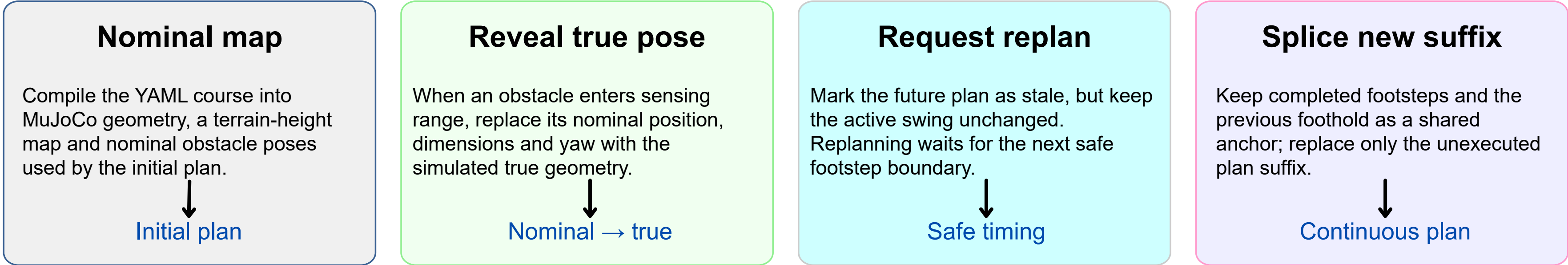
Level 1 · fixed, fully known → **Level 2 · 1-D position uncertainty** → **Level 3 · 2-D position + heading uncertainty**

Pipeline



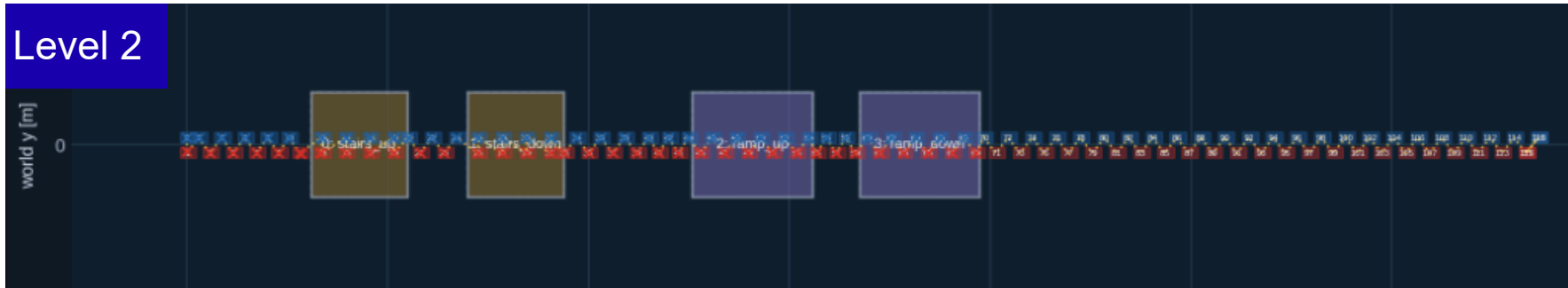
Methodology

Terrain sensing and online replanning

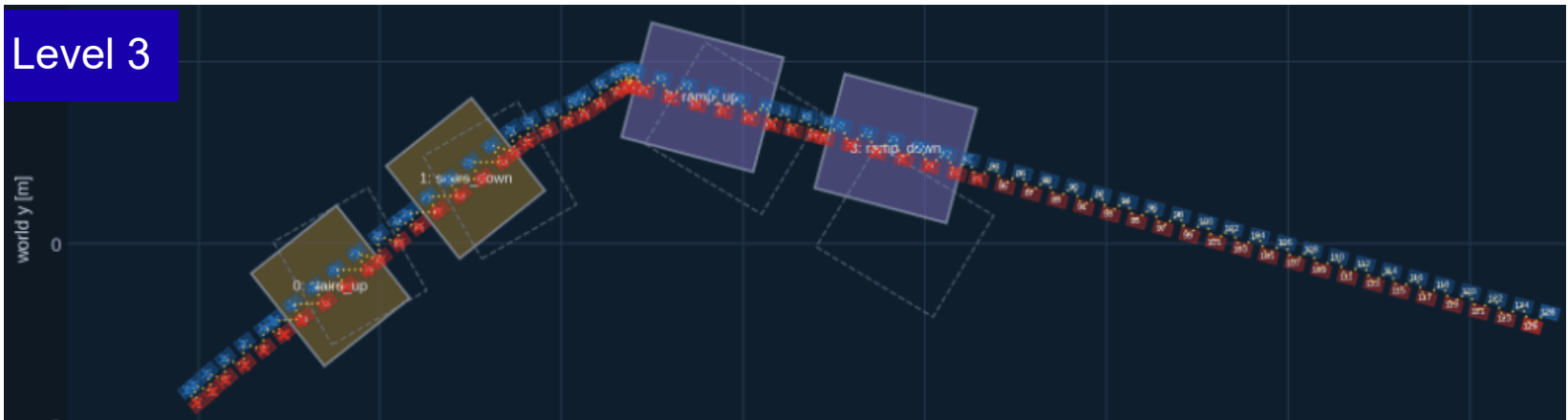


Online replanning: perception changes the map, not the currently executing swing.

nominal map → true-pose reveal → pending replan → step boundary → new suffix



1-D online terrain reveal — straight-corridor replanning

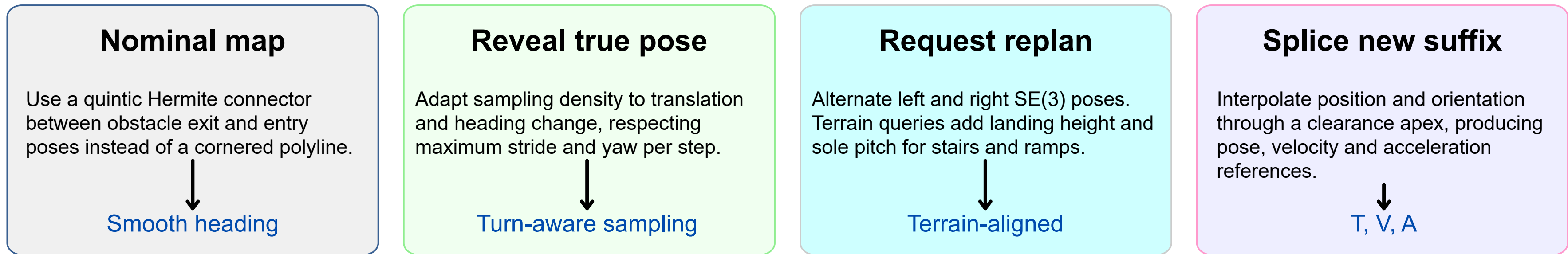


2-D pose-aware replanning — position and heading updates

Replanning output: an updated foothold sequence with a continuous old-to-new seam

Methodology

Path and motion planning



Obstacle-to-obstacle trajectory: a fifth-order curve uses positions p_0, p_1 , and heading-aligned tangents v_0, v_1 .

$$p(t) = h_{00}p_0 + h_{10}v_0 + h_{01}p_1 + h_{11}v_1$$

Start position

departure direction

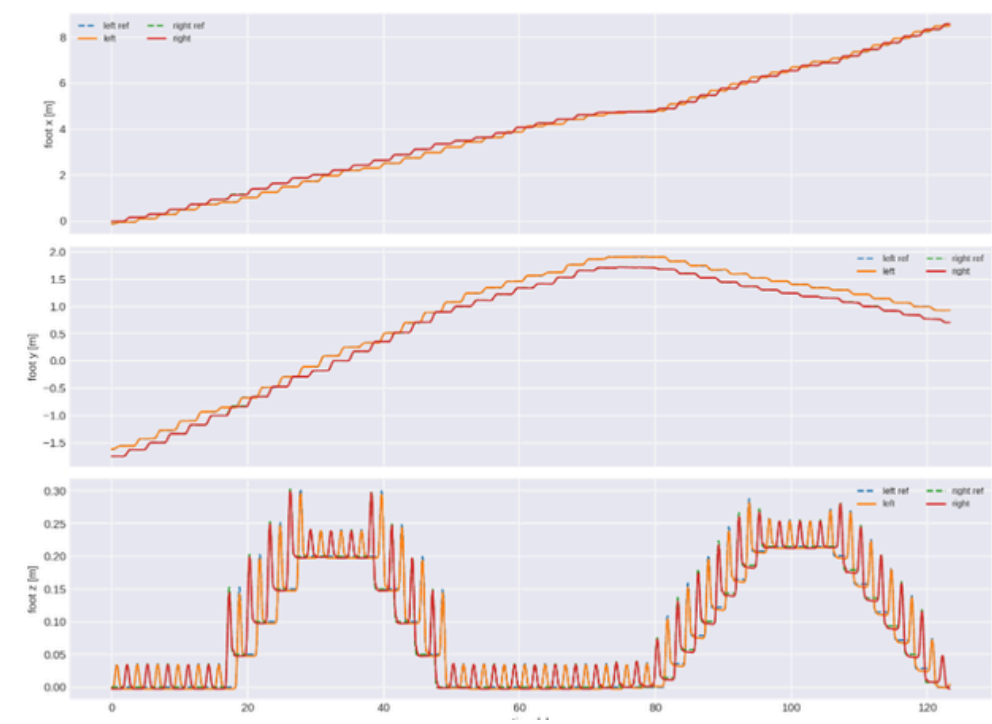
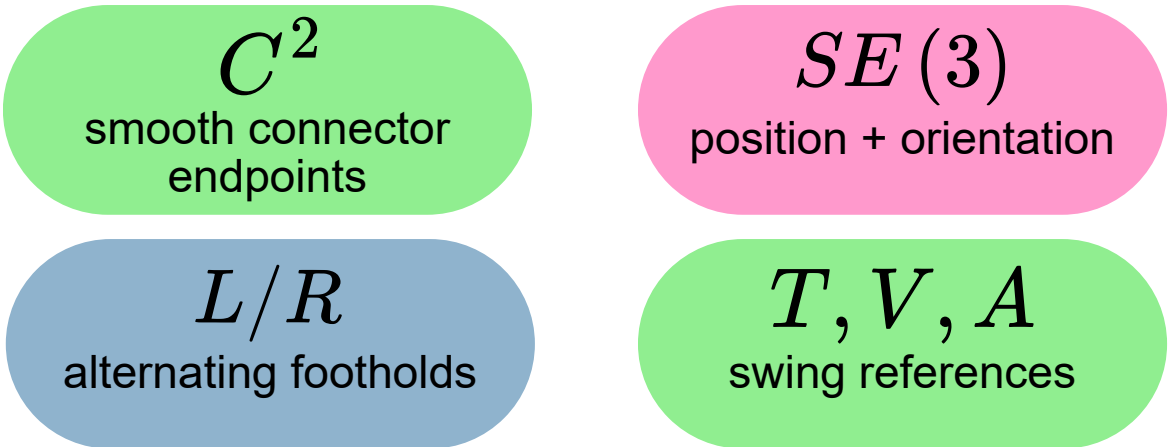
End position

Arrival direction

$$m = \frac{\Delta s}{s_{max}} + \frac{|\Delta \psi|}{\psi_{max}}$$

From geometric path to executable motion

- Match obstacle positions and headings
- Resample under stride and yaw bounds
- Offset alternating feet from the centreline
- Query terrain height and pitch
- Interpolate swing-foot SE(3) motion



Level 3 foot-trajectory execution — planned and measured foot motion

Planning output: a smooth route, reachable footholds and continuous swing-foot references

Methodology

LIP-MPC Balance Control

1 Read CoM state

Blend measured CoM position and velocity with the predicted LIP state.

2 Predict CoM motion

Use exact discrete LIP dynamics to preview the next 3.0 s.

3 Optimise ZMP

Track the planned ZMP, smooth CoM velocity and constrain ZMP inside the yawed sole.

4 Apply & adapt step

Apply only u_0 . Lateral DCM error shifts the next foothold by at most ± 6 cm.

Linear inverted-pendulum model

$$\ddot{c}_{xy} = \omega^2 (c_{xy} - z_{mp}) \cdot \omega = \sqrt{(g / h)}$$

Horizontal balance comes from ZMP control; vertical CoM height follows the terrain separately.

$$c_{z,ref} = h_{nom} + h_{terrain}(c_x + 0.08 \text{ m})$$

Measured CoM x_k

Yaw-aware MPC

ZMP u_0 + CoM refs



1 kHz
LIP interpolation

20 Hz
MPC update · $\Delta t = 0.05$ s

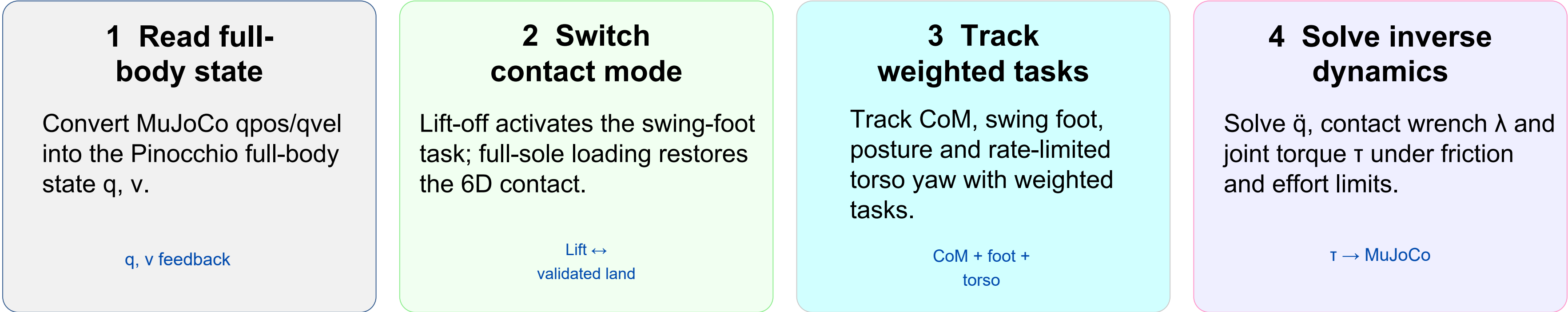
± 6 cm
DCM footstep shift

60
Samples · 2 footsteps

Output: feasible ZMP u_0 and smooth CoM position, velocity and acceleration references for TSID

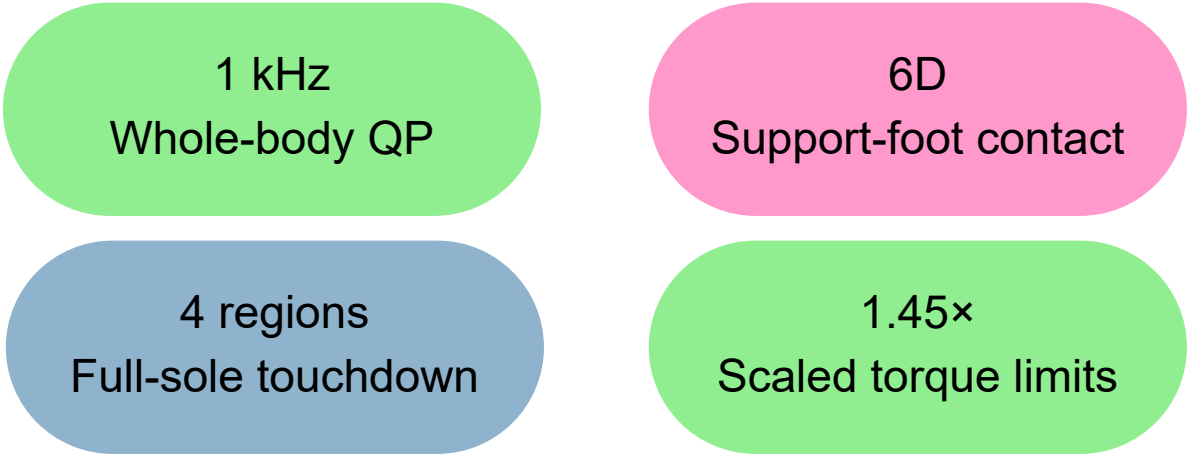
Methodology

TSID Whole-Body Control



Whole-body inverse dynamics

$M(q)\ddot{q} + h(q,\dot{q}) = S^T\tau + Jc^T\lambda$
TSID coordinates motion and contact tasks; an invalid solve falls back to the previous valid torque.



Output: bounded joint torques for MuJoCo with validated touchdown and safe QP fallback

Experimental Results

Level 1 · 100%
20/20 runs completed

Level 2 · 100%
20/20 runs completed

Level 3 · ~100%
19/20 successful runs

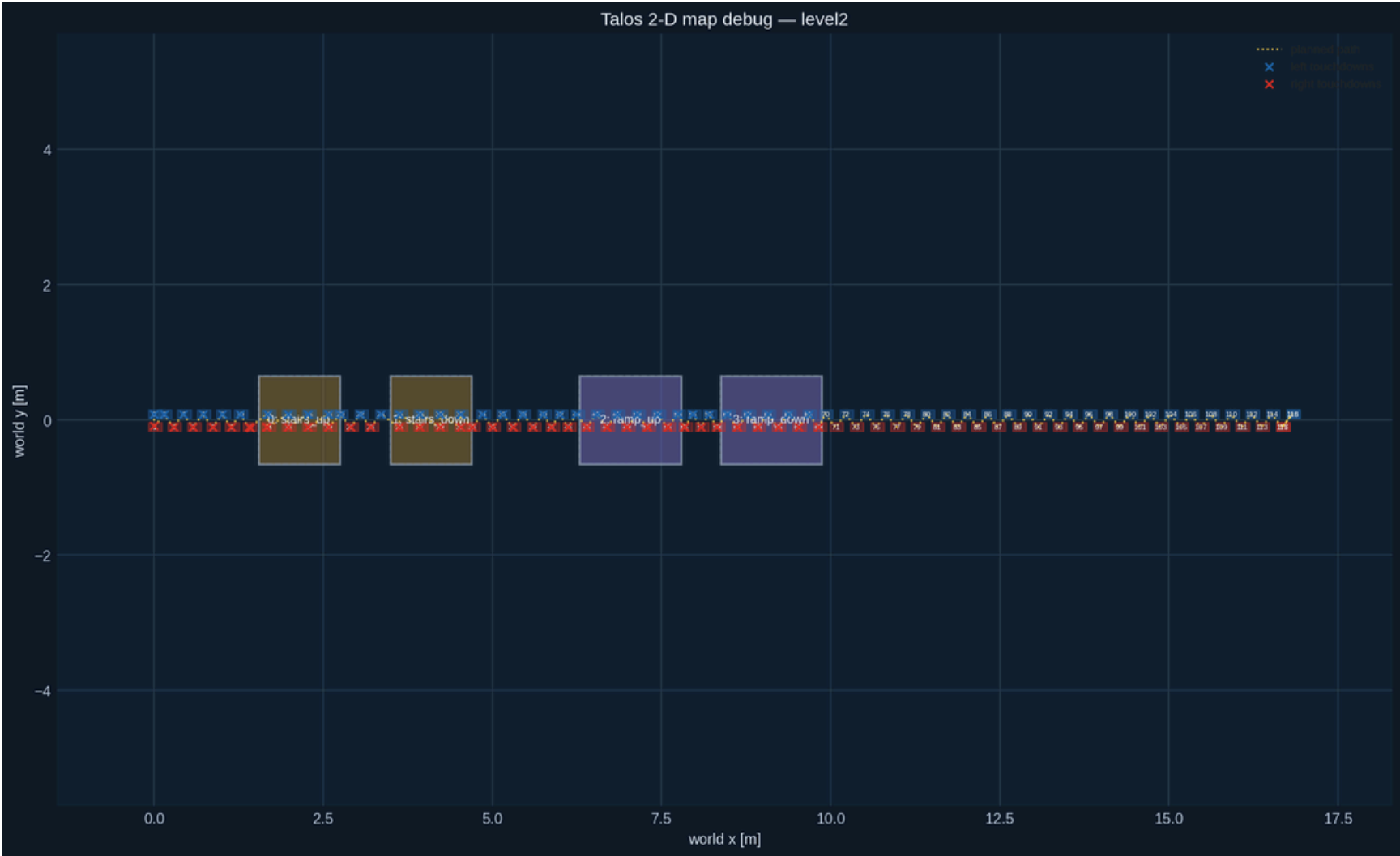
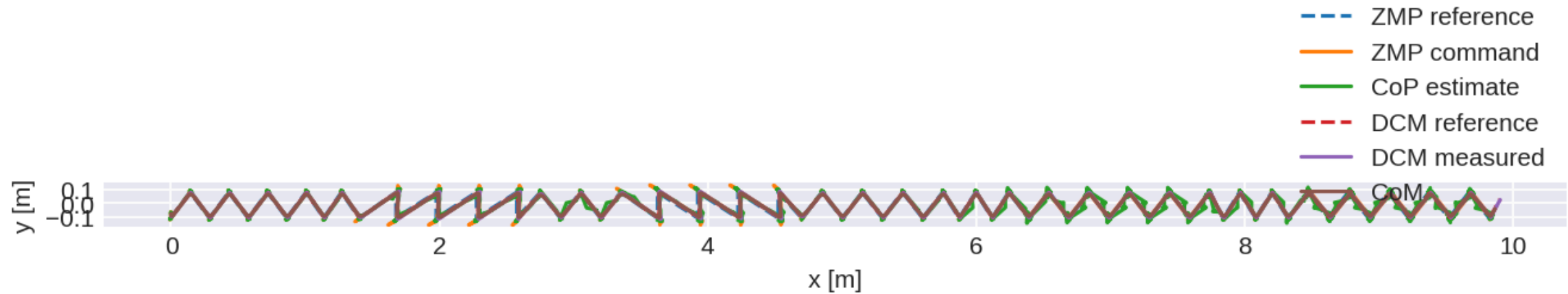


Fig. Top-down obstacle track in level 2

Fig. ZMP / DCM tracking in level 2



Experimental Results

Level 1 · 100%
20/20 runs completed

Level 2 · 100%
20/20 runs completed

Level 3 · ~100%
19/20 successful runs

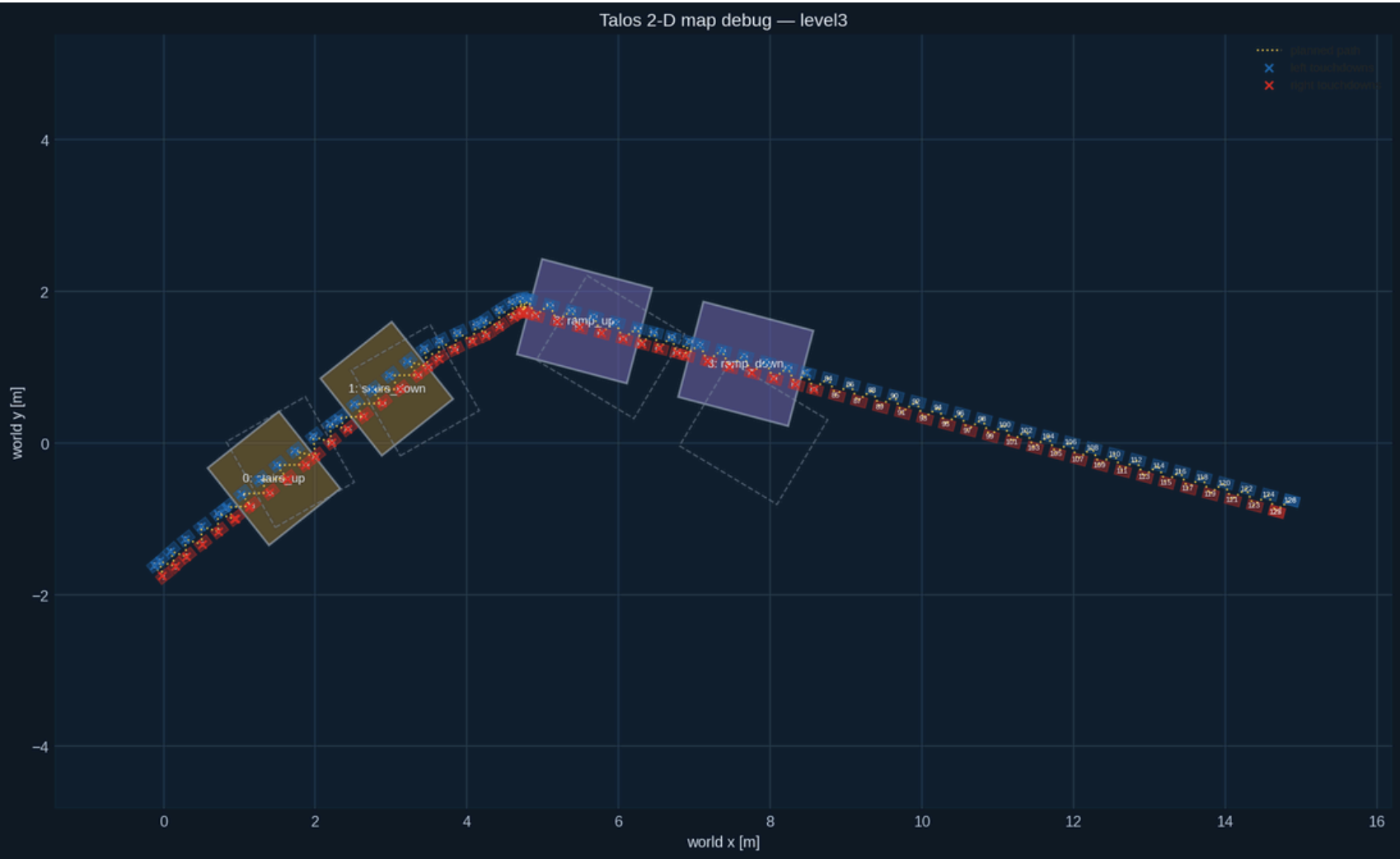
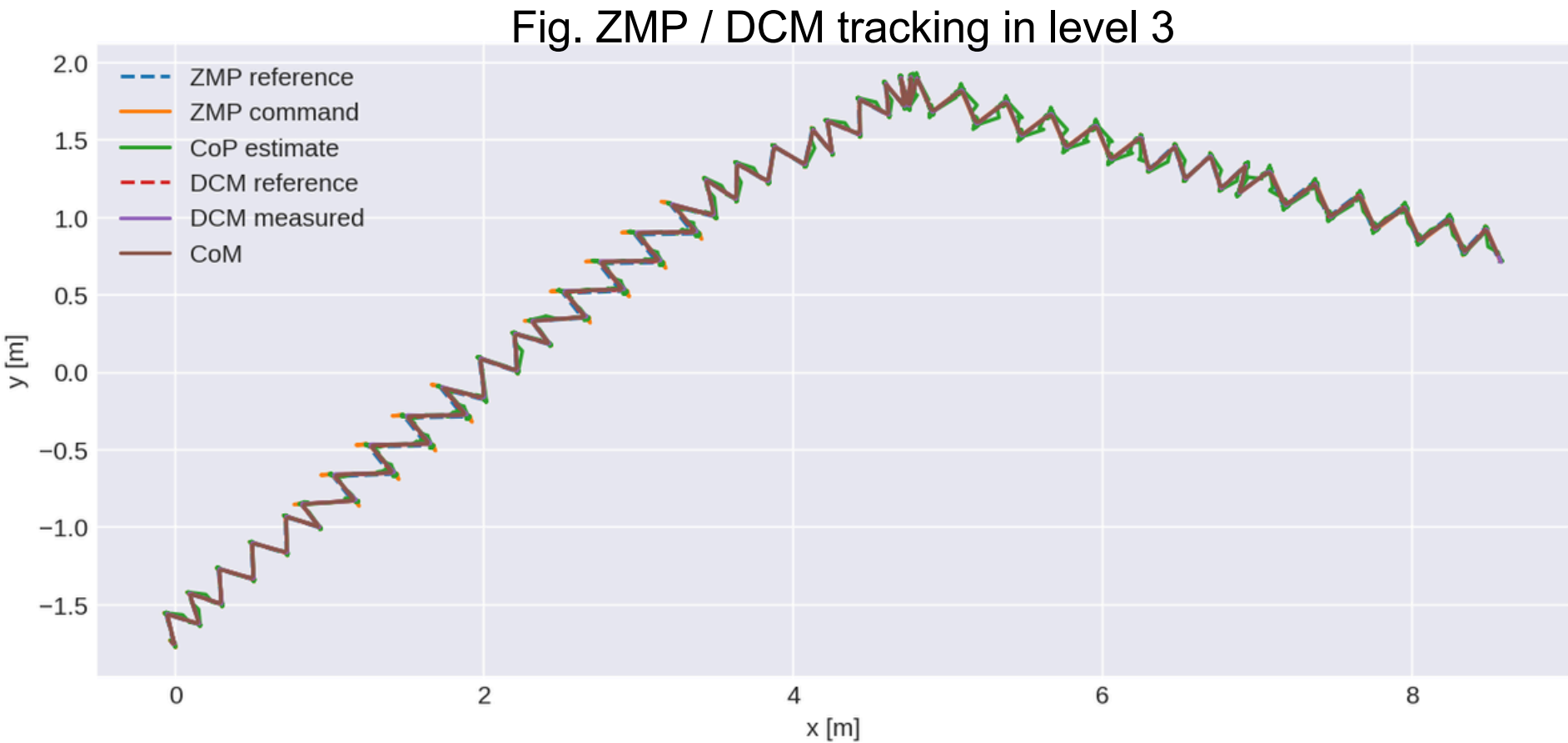
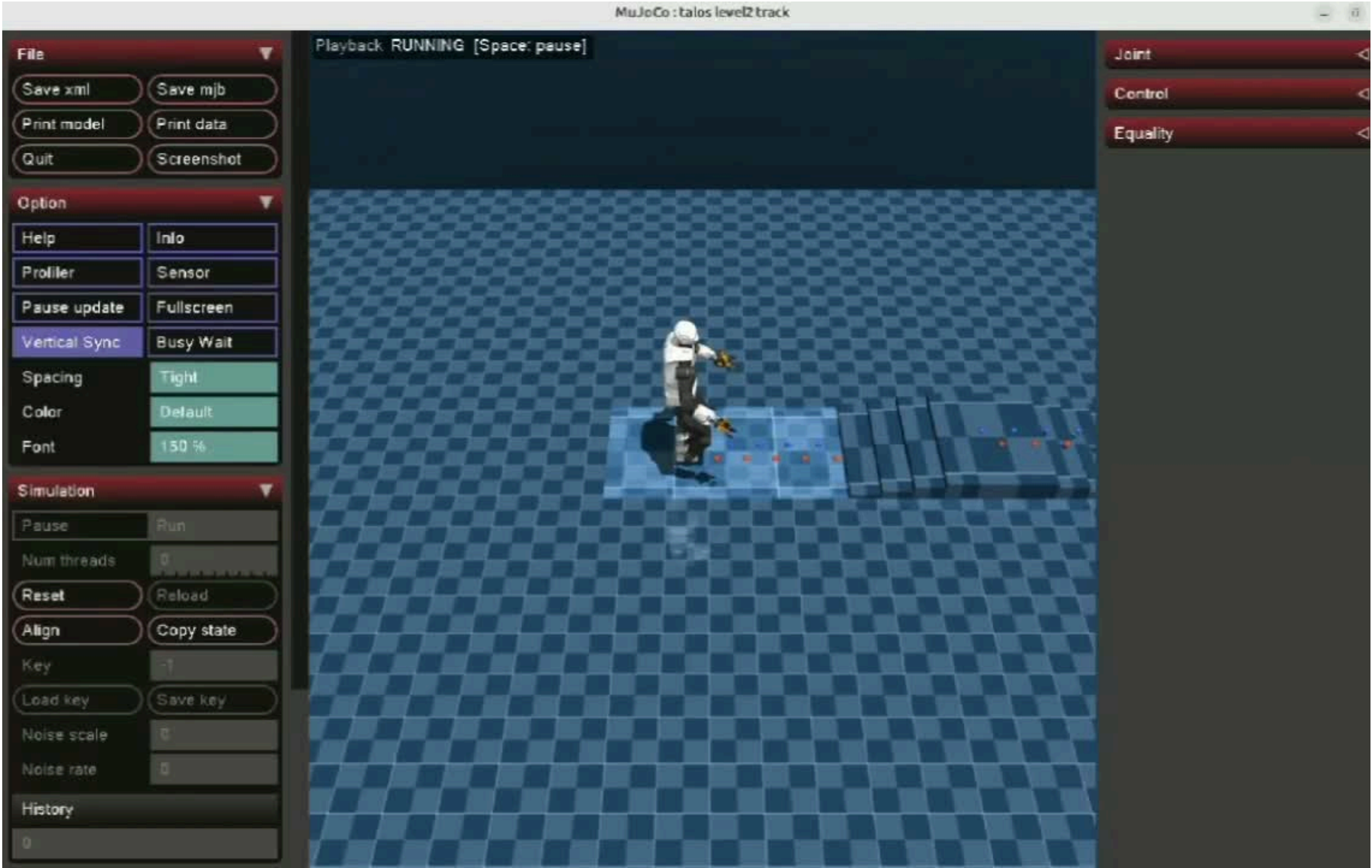


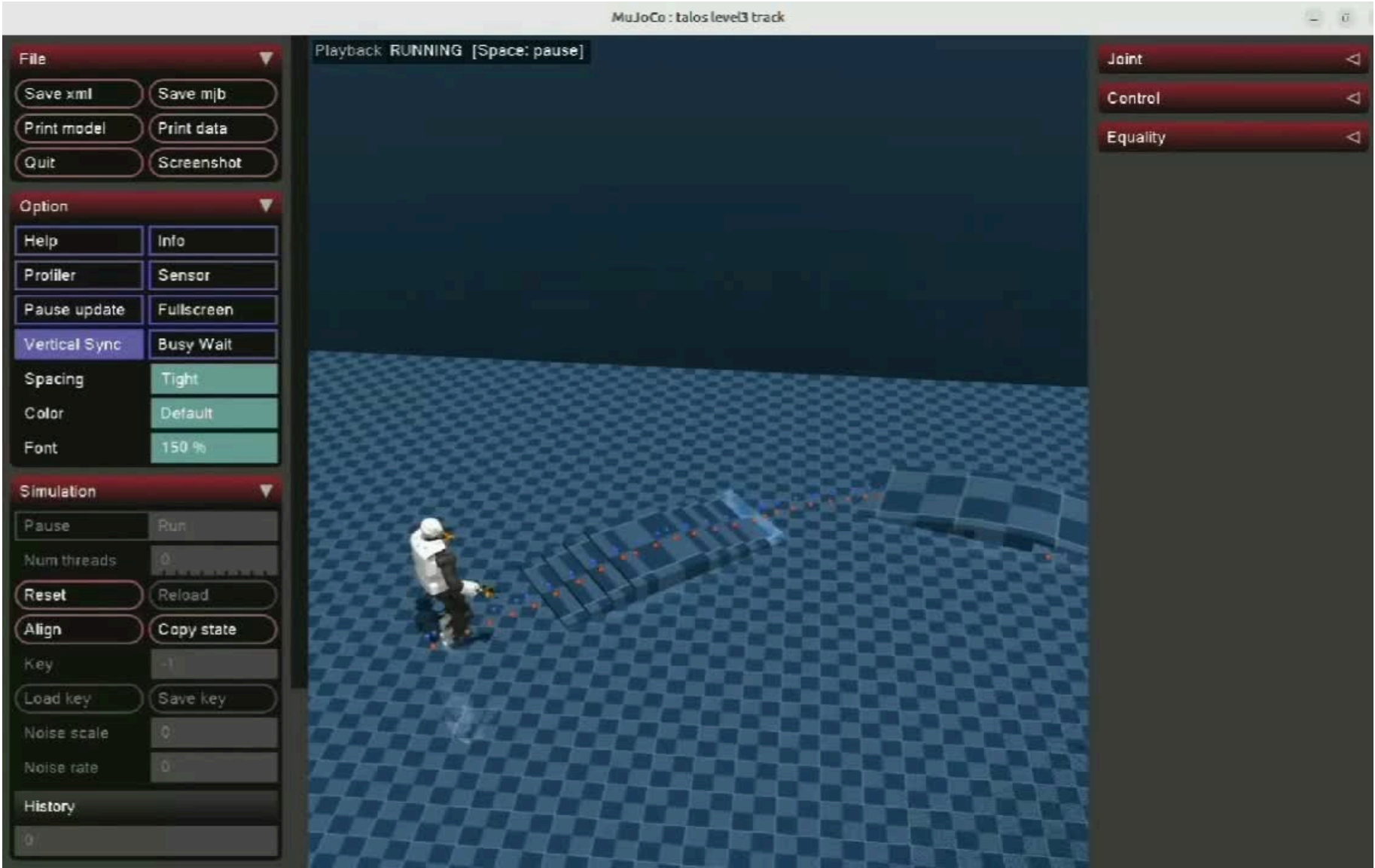
Fig. Top-down obstacle track in level 3



Demo



Level 2 example



Level 3 example

Thank you